Multisensory Fitness, Inc.

EMC TEST REPORT FOR

SMARTfit ProTrainer 3
Model: 1-20203

Tested to The Following Standards:

FCC Part 15 Subpart B Section 15.107 & 15.109

Report No.: 97844-5

Date of issue: July 15, 2016

ACCREDITED
Testing Certificates:
803.01, 803.02, 803.05, 803.06

This test report bears the accreditation symbol indicating that the testing performed herein meets the test and reporting requirements of ISO/IEC 17025 under the applicable scope of EMC testing for CKC Laboratories, Inc.

We strive to create long-term, trust based relationships by providing sound, adaptive, customer first testing services. We embrace each of our customers' unique EMC challenges, not as an interruption to set processes, but rather as the reason we are in business.

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ADMINISTRATIVE INFORMATION

Test Report Information

REPORT PREPARED FOR:

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REPORT PREPARED BY:

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Project Number: 97844

DATE OF EQUIPMENT RECEIPT: July 8, 2016
DATE(S) OF TESTING: July 8, 2016

Report Authorization

The test data contained in this report documents the observed testing parameters pertaining to and are relevant for only the sample equipment tested in the agreed upon operational mode(s) and configuration(s) as identified herein. Compliance assessment remains the client's responsibility. This report may not be used to claim product endorsement by A2LA or any government agencies. This test report has been authorized for release under quality control from CKC Laboratories, Inc.

Steve Behm
Director of Quality Assurance & Engineering Services
CKC Laboratories, Inc.

Test Facility Information



Our laboratories are configured to effectively test a wide variety of product types. CKC utilizes first class test equipment, anechoic chambers, data acquisition and information services to create accurate, repeatable and affordable test results.

TEST LOCATION(S):
CKC Laboratories, Inc.
110 Olinda Place
Brea, CA 92823

Software Versions

CKC Laboratories Proprietary Software	Version
EMITest Emissions	5.03.02
EMITest Immunity	5.03.02

Site Registration & Accreditation Information

Location	CB #	TAIWAN	CANADA	FCC	JAPAN
Brea D	US0060	SL2-IN-E-1146R	3082D-2	100638	A-0147

SUMMARY OF RESULTS

Standard / Specification: FCC Part 15 Subpart B

Test Procedure	Description	Modifications	Results
15.107 Class B	Conducted Emissions	Mod. #1	Pass
15.109 Class B	Radiated Emissions	Mod. #1	Pass

NA = Not Applicable

Modifications During Testing

This list is a summary of the modifications made to the equipment during testing.

Summary of Conditions
Mod #1: Grounded the shield of the three panel cables internal to the controller by using a single wire from the power supply ground terminal. Grounded the dc minus using the same single wire from the power supply ground terminal. Removed the approximately 100mm length extension cable from the top six LED boards.

Modifications listed above must be incorporated into all production units.

Conditions During Testing

This list is a summary of the conditions noted to the equipment during testing.

Summary of Conditions
None

EQUIPMENT UNDER TEST (EUT)

During testing numerous configurations may have been utilized. The configurations listed below support compliance to the standard(s) listed in the Summary of Results section.

Configuration 1

Equipment Tested:

Device	Manufacturer	Model #	S/N
SMARTfit ProTrainer 3	Multisensory Fitness, Inc.	1-20203	None

Support Equipment:

Device	Manufacturer	Model #	S/N
None			

FCC PART 15 SUBPART B

15.107 AC Conducted Emissions

Test Notes: Conducted Disturbances at Mains Terminals, LISN method.

Test Setup / Conditions / Data

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**15.107 AC Mains Class B - Quasi-peak**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Conducted Emissions**	Time:	15:15:55
Tested by:	S. Yamamoto	Sequence#:	3
Software:	EMITest 5.03.02		120V 60Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 1			

Support Equipment:

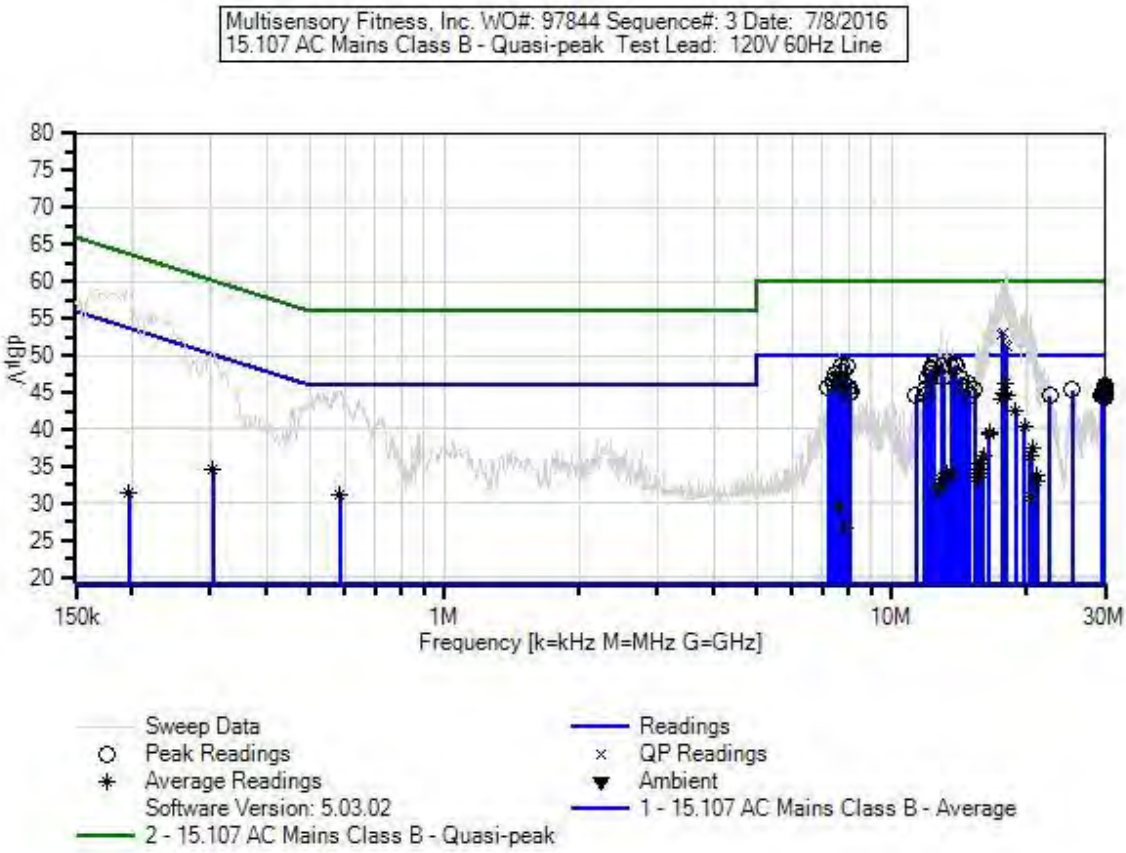
Device	Manufacturer	Model #	S/N
Configuration 1			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 150kHz to 30MHz. RBW=VBW=9kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
Test Method: ANSI C63.4 (2014).
Modification 1 was in place during testing.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 3 Date: 7/8/2016
15.107 AC Mains Class B - Quasi-peak Test Lead: 120V 60Hz Line
dBµV
80
75
70
65
60
55
50
45
40
35
30
25
20
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Peak Readings
Average Readings
Software Version: 5.03.02
2 - 15.107 AC Mains Class B - Quasi-peak
Readings
QP Readings
Ambient
1 - 15.107 AC Mains Class B - Average

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T1	AN02343	High Pass Filter	HE9615-150K-50-720B	1/8/2015	1/8/2017
T2	ANP01910	Cable	RG-142	11/30/2015	11/30/2017
T3	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T4	ANP06085	Attenuator	SA18N10W-09	12/17/2014	12/17/2016
T5	AN00969A	50uH LISN-Line 1 (L1) (dB)	3816/2NM	3/12/2015	3/12/2017
	AN00969A	50uH LISN-Line 2 (L2) (dB)	3816/2NM	3/12/2015	3/12/2017

Measurement Data: Reading listed by margin. Test Lead: Line

#	Freq	Rdng	T1 T5	T2	T3	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV	dBµV	dB	Ant
1	12.444M	42.3	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.8	50.0	-1.2	Line
2	13.834M	42.3	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.8	50.0	-1.2	Line
3	7.716M	42.5	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	48.7	50.0	-1.3	Line
4	7.942M	42.5	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	48.7	50.0	-1.3	Line
5	13.924M	42.1	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.6	50.0	-1.4	Line
6	13.617M	42.0	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.5	50.0	-1.5	Line
7	13.003M	42.0	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.5	50.0	-1.5	Line
8	12.264M	41.7	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.2	50.0	-1.8	Line
9	12.300M	41.7	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.2	50.0	-1.8	Line
10	12.228M	41.5	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	48.0	50.0	-2.0	Line
11	7.788M	41.6	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	47.8	50.0	-2.2	Line
12	12.209M	41.1	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.6	50.0	-2.4	Line
13	7.455M	41.5	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	47.6	50.0	-2.4	Line
14	14.312M	40.4	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	47.1	50.0	-2.9	Line
15	11.984M	40.7	+0.1 +0.3	+0.1	+0.2	+5.7	+0.0	47.1	50.0	-2.9	Line
16	13.067M	40.5	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.0	50.0	-3.0	Line
17	14.203M	40.2	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	46.9	50.0	-3.1	Line

18	7.545M	40.6	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	46.7	50.0	-3.3	Line
19	7.391M	40.5	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	46.6	50.0	-3.4	Line
20	7.608M	40.3	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	46.4	50.0	-3.6	Line
21	7.834M	40.1	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	46.3	50.0	-3.7	Line
22	7.419M	40.2	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	46.3	50.0	-3.7	Line
23	14.781M	39.6	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	46.3	50.0	-3.7	Line
24	17.939M Ave	39.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	46.1	50.0	-3.9	Line
25	29.982M	38.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.9	50.0	-4.1	Line
26	8.041M	39.5	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	45.7	50.0	-4.3	Line
27	7.229M	39.5	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	45.6	50.0	-4.4	Line
28	14.573M	38.9	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	45.6	50.0	-4.4	Line
29	29.916M	38.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.6	50.0	-4.4	Line
30	29.955M	38.0	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.5	50.0	-4.5	Line
31	15.304M	38.7	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	45.4	50.0	-4.6	Line
32	25.409M	38.2	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	45.4	50.0	-4.6	Line
33	29.939M	37.9	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.4	50.0	-4.6	Line
34	11.921M	38.9	+0.1 +0.3	+0.1	+0.2	+5.7	+0.0	45.3	50.0	-4.7	Line
35	29.868M	37.7	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.2	50.0	-4.8	Line
36	29.771M	37.6	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.1	50.0	-4.9	Line
37	8.104M	38.9	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	45.1	50.0	-4.9	Line
38	29.975M	37.6	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.1	50.0	-4.9	Line
39	29.913M	37.6	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.1	50.0	-4.9	Line
40	29.907M	37.5	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	45.0	50.0	-5.0	Line
41	29.751M	37.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.9	50.0	-5.1	Line
42	8.077M	38.7	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	44.9	50.0	-5.1	Line
43	29.788M	37.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.9	50.0	-5.1	Line

44	29.991M	37.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.9	50.0	-5.1	Line
45	29.829M	37.4	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.9	50.0	-5.1	Line
46	17.776M Ave	38.0	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	44.8	50.0	-5.2	Line
47	29.768M	37.3	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.8	50.0	-5.2	Line
48	29.848M	37.3	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.8	50.0	-5.2	Line
49	11.858M	38.3	+0.1 +0.3	+0.1	+0.2	+5.7	+0.0	44.7	50.0	-5.3	Line
50	11.343M	38.3	+0.1 +0.3	+0.1	+0.2	+5.7	+0.0	44.7	50.0	-5.3	Line
51	29.999M	37.2	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.7	50.0	-5.3	Line
52	29.859M	37.2	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.7	50.0	-5.3	Line
53	29.497M	37.2	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.7	50.0	-5.3	Line
54	22.603M	37.4	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	44.6	50.0	-5.4	Line
55	14.961M	37.9	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	44.6	50.0	-5.4	Line
56	29.892M	37.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.6	50.0	-5.4	Line
57	29.876M	37.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.6	50.0	-5.4	Line
58	29.781M	37.1	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.6	50.0	-5.4	Line
59	18.074M Ave	37.7	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	44.5	50.0	-5.5	Line
60	29.992M	37.0	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.5	50.0	-5.5	Line
61	29.865M	37.0	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.5	50.0	-5.5	Line
62	29.918M	36.9	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.4	50.0	-5.6	Line
63	29.928M	36.9	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.4	50.0	-5.6	Line
64	29.996M	36.8	+0.2 +0.9	+0.2	+0.5	+5.7	+0.0	44.3	50.0	-5.7	Line
65	17.650M Ave	37.4	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	44.2	50.0	-5.8	Line

66	17.776M QP	46.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	53.1	60.0	-6.9	Line
^	17.776M	52.8	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	59.6	50.0 see average and quasi peak data	+9.6	Line
68	17.650M QP	46.0	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	52.8	60.0	-7.2	Line
^	17.650M	53.2	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	60.0	50.0 see average and quasi peak data	+10.0	Line
70	18.940M Ave	35.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	42.5	50.0	-7.5	Line
^	18.940M	49.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	56.5	50.0 see average data	+6.5	Line
72	17.939M QP	44.6	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	51.4	60.0	-8.6	Line
^	17.939M	53.8	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	60.6	50.0 see average and quasi peak data	+10.6	Line
74	18.074M QP	44.4	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	51.2	60.0	-8.8	Line
^	18.074M	54.1	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	60.9	50.0 see average and quasi peak data	+10.9	Line
76	19.797M Ave	33.5	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	40.5	50.0	-9.5	Line
^	19.797M	48.8	+0.2 +0.6	+0.1	+0.4	+5.7	+0.0	55.8	50.0 see average data	+5.8	Line
78	16.576M Ave	32.9	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	39.7	50.0	-10.3	Line
^	16.576M	49.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	56.1	50.0 see average data	+6.1	Line
80	16.504M Ave	32.5	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	39.3	50.0	-10.7	Line
^	16.504M	48.1	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	54.9	50.0 see average data	+4.9	Line
82	20.618M Ave	30.4	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	37.5	50.0	-12.5	Line
^	20.618M	46.2	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	53.3	50.0 see average data	+3.3	Line
84	16.062M Ave	29.6	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	36.4	50.0	-13.6	Line
^	16.062M	46.2	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	53.0	50.0 see average data	+3.0	Line
86	20.428M Ave	29.2	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	36.3	50.0	-13.7	Line
^	20.429M	48.1	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	55.2	50.0 see average data	+5.2	Line
88	15.900M Ave	29.0	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	35.8	50.0	-14.2	Line
^	15.900M	44.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	51.1	50.0 see average data	+1.1	Line

90 Ave	15.755M	28.5	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	35.3	50.0	-14.7	Line
^	15.755M	43.7	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	50.5	50.0 see average data	+0.5	Line
92 Ave	584.506k	25.2	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	31.2	46.0	-14.8	Line
^	584.505k	39.3	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	45.3	46.0 see average data	-0.7	Line
94 Ave	15.719M	27.7	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	34.5	50.0	-15.5	Line
^	15.719M	42.7	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	49.5	50.0 see average data	-0.5	Line
96 Ave	304.531k	28.6	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	34.5	50.1	-15.6	Line
^	304.531k	44.7	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	50.6	50.1 see average data	+0.5	Line
98 Ave	13.310M	27.6	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	34.1	50.0	-15.9	Line
^	13.310M	45.7	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	52.2	50.0 see average data	+2.2	Line
100 Ave	15.575M	27.2	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	34.0	50.0	-16.0	Line
^	15.575M	43.0	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	49.8	50.0 see average data	-0.2	Line
102 Ave	13.752M	27.5	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	34.0	50.0	-16.0	Line
^	13.752M	44.4	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	50.9	50.0 see average data	+0.9	Line
104 Ave	20.997M	26.6	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	33.8	50.0	-16.2	Line
^	20.997M	44.3	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	51.5	50.0 see average data	+1.5	Line
106 Ave	15.539M	26.9	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	33.6	50.0	-16.4	Line
^	15.539M	45.2	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	51.9	50.0 see average data	+1.9	Line
108 Ave	15.467M	26.8	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	33.5	50.0	-16.5	Line
^	15.467M	43.5	+0.2 +0.4	+0.1	+0.3	+5.7	+0.0	50.2	50.0 see average data	+0.2	Line
110 Ave	12.886M	26.8	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	33.3	50.0	-16.7	Line
^	12.886M	47.2	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	53.7	50.0 see average data	+3.7	Line
112 Ave	21.115M	25.7	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	32.9	50.0	-17.1	Line
^	21.115M	43.8	+0.2 +0.7	+0.2	+0.4	+5.7	+0.0	51.0	50.0 see average data	+1.0	Line
114 Ave	12.832M	26.1	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	32.6	50.0	-17.4	Line
^	12.832M	43.4	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	49.9	50.0 see average data	-0.1	Line

116	12.760M Ave	25.6	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	32.1	50.0	-17.9	Line
^	12.760M	44.7	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	51.2	50.0 see average data	+1.2	Line
118	12.805M Ave	25.5	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	32.0	50.0	-18.0	Line
^	12.805M	44.1	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	50.6	50.0 see average data	+0.6	Line
120	20.492M Ave	23.8	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	30.9	50.0	-19.1	Line
^	20.492M	47.6	+0.2 +0.7	+0.1	+0.4	+5.7	+0.0	54.7	50.0 see average data	+4.7	Line
122	7.671M Ave	23.4	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	29.5	50.0	-20.5	Line
^	7.671M	44.3	+0.1 +0.2	+0.0	+0.1	+5.7	+0.0	50.4	50.0 see average data	+0.4	Line
124	197.268k Ave	25.5	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	31.4	53.7	-22.3	Line
^	197.268k	52.7	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	58.6	53.7 see average data	+4.9	Line
126	7.879M Ave	20.4	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	26.6	50.0	-23.4	Line
^	7.879M	43.1	+0.1 +0.3	+0.0	+0.1	+5.7	+0.0	49.3	50.0 see average data	-0.7	Line

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**15.107 AC Mains Class B - Quasi-peak**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Conducted Emissions**	Time:	15:36:14
Tested by:	S. Yamamoto	Sequence#:	4
Software:	EMITest 5.03.02		120V 60Hz

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 1			

Support Equipment:

Device	Manufacturer	Model #	S/N
Configuration 1			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 150kHz to 30MHz. RBW=VBW=9kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
Test Method: ANSI C63.4 (2014).
Modification 1 was in place during test.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 4 Date: 7/8/2016
15.107 AC Mains Class B - Quasi-peak Test Lead: 120V 60Hz Neutral
dBµV
80
75
70
65
60
55
50
45
40
35
30
25
20
150k
1M
10M
30M
Frequency [k=kHz M=MHz G=GHz]
Sweep Data
Peak Readings
Average Readings
Software Version: 5.03.02
2 - 15.107 AC Mains Class B - Quasi-peak
Readings
QP Readings
Ambient
1 - 15.107 AC Mains Class B - Average

Test Equipment:

ID	Asset #	Description	Model	Calibration Date	Cal Due Date
	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T1	AN02343	High Pass Filter	HE9615-150K-50-720B	1/8/2015	1/8/2017
T2	ANP01910	Cable	RG-142	11/30/2015	11/30/2017
T3	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T4	ANP06085	Attenuator	SA18N10W-09	12/17/2014	12/17/2016
	AN00969A	50uH LISN-Line 1 (L1) (dB)	3816/2NM	3/12/2015	3/12/2017
T5	AN00969A	50uH LISN-Line 2 (L2) (dB)	3816/2NM	3/12/2015	3/12/2017

Measurement Data: Reading listed by margin. Test Lead: Neutral

#	Freq	Rdng	T1 T5	T2	T3	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV	dBµV	dB	Ant
1	631.773k	38.9	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	44.9	46.0	-1.1	Neutr
2	15.891M	42.0	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	48.9	50.0	-1.1	Neutr
3	15.963M	41.9	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	48.8	50.0	-1.2	Neutr
4	15.800M	41.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	48.7	50.0	-1.3	Neutr
5	29.460M	41.0	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	48.6	50.0	-1.4	Neutr
6	28.829M	40.9	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	48.5	50.0	-1.5	Neutr
7	639.046k	38.4	+0.2 +0.1	+0.0	+0.0	+5.7	+0.0	44.4	46.0	-1.6	Neutr
8	27.187M	40.6	+0.2 +1.0	+0.2	+0.4	+5.7	+0.0	48.1	50.0	-1.9	Neutr
9	28.883M	40.5	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	48.1	50.0	-1.9	Neutr
10	15.918M	41.1	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	48.0	50.0	-2.0	Neutr
11	26.889M	40.6	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	48.0	50.0	-2.0	Neutr
12	12.877M	41.4	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.9	50.0	-2.1	Neutr
13	28.685M	40.2	+0.2 +1.0	+0.2	+0.5	+5.7	+0.0	47.8	50.0	-2.2	Neutr
14	12.940M	41.2	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	47.7	50.0	-2.3	Neutr
15	26.366M	40.3	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	47.7	50.0	-2.3	Neutr
16	13.599M	40.9	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	47.5	50.0	-2.5	Neutr
17	26.438M	40.0	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	47.4	50.0	-2.6	Neutr

18	13.076M	40.4	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.9	50.0	-3.1	Neutr
19	23.930M	39.4	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	46.8	50.0	-3.2	Neutr
20	12.661M	40.2	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.7	50.0	-3.3	Neutr
21	12.778M	40.2	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.7	50.0	-3.3	Neutr
22	13.536M	39.6	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	46.2	50.0	-3.8	Neutr
23	13.770M	39.5	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	46.1	50.0	-3.9	Neutr
24	20.185M	38.9	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	46.1	50.0	-3.9	Neutr
25	12.724M	39.5	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	46.0	50.0	-4.0	Neutr
26	12.327M	39.4	+0.1 +0.4	+0.1	+0.2	+5.7	+0.0	45.9	50.0	-4.1	Neutr
27	13.843M	39.1	+0.1 +0.5	+0.1	+0.2	+5.7	+0.0	45.7	50.0	-4.3	Neutr
28	20.366M	38.5	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	45.7	50.0	-4.3	Neutr
29	20.321M	38.2	+0.2 +0.8	+0.1	+0.4	+5.7	+0.0	45.4	50.0	-4.6	Neutr
30	839.027k	35.3	+0.1 +0.1	+0.0	+0.0	+5.7	+0.0	41.2	46.0	-4.8	Neutr
31	15.313M	38.3	+0.2 +0.5	+0.1	+0.3	+5.7	+0.0	45.1	50.0	-4.9	Neutr
32	17.533M Ave	36.7	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	43.6	50.0	-6.4	Neutr
^	17.533M	49.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	56.3	50.0 see average data	+6.3	Neutr
34	17.470M Ave	36.7	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	43.6	50.0	-6.4	Neutr
^	17.470M	49.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	56.7	50.0 see average data	+6.7	Neutr
36	18.164M QP	46.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	53.3	60.0	-6.7	Neutr
37	17.632M QP	46.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	53.3	60.0	-6.7	Neutr
38	18.200M QP	45.9	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	52.8	60.0	-7.2	Neutr
39	17.632M Ave	35.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	42.7	50.0	-7.3	Neutr
^	17.632M	52.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	59.3	50.0 see average and quasi peak data	+9.3	Neutr
41	18.318M Ave	35.4	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	42.4	50.0	-7.6	Neutr
^	18.318M	49.0	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	56.0	50.0 see average data	+6.0	Neutr

43	17.235M Ave	35.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	42.3	50.0	-7.7	Neutr
44	17.235M Ave	35.3	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	42.2	50.0	-7.8	Neutr
^	17.235M	50.0	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	56.9	50.0 see average data	+6.9	Neutr
46	17.921M Ave	35.3	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	42.2	50.0	-7.8	Neutr
^	17.921M	50.5	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.4	50.0 see average data	+7.4	Neutr
48	18.101M Ave	34.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	41.7	50.0	-8.3	Neutr
^	18.101M	51.0	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	57.9	50.0 see average data	+7.9	Neutr
50	18.164M Ave	34.6	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	41.5	50.0	-8.5	Neutr
^	18.164M	52.8	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	59.7	50.0 see average and quasi peak data	+9.7	Neutr
52	18.534M Ave	34.3	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	41.3	50.0	-8.7	Neutr
^	18.534M	49.5	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	56.5	50.0 see average data	+6.5	Neutr
54	18.200M Ave	34.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	41.3	50.0	-8.7	Neutr
^	18.200M	53.2	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	60.1	50.0 see average and quasi peak data	+10.1	Neutr
56	18.372M Ave	33.6	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	40.6	50.0	-9.4	Neutr
^	18.372M	51.9	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	58.9	50.0 see average data	+8.9	Neutr
58	16.847M Ave	33.4	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	40.3	50.0	-9.7	Neutr
^	16.847M	46.1	+0.2 +0.6	+0.1	+0.3	+5.7	+0.0	53.0	50.0 see average data	+3.0	Neutr
60	18.679M Ave	33.0	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	40.0	50.0	-10.0	Neutr
^	18.679M	47.8	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	54.8	50.0 see average data	+4.8	Neutr
62	25.518M Ave	32.2	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	39.6	50.0	-10.4	Neutr
^	25.518M	46.7	+0.2 +0.9	+0.2	+0.4	+5.7	+0.0	54.1	50.0 see average data	+4.1	Neutr
64	18.796M Ave	32.5	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	39.5	50.0	-10.5	Neutr
^	18.796M	46.8	+0.2 +0.7	+0.1	+0.3	+5.7	+0.0	53.8	50.0 see average data	+3.8	Neutr

66	29.984M	31.3	+0.2	+0.2	+0.5	+5.7	+0.0	39.0	50.0	-11.0	Neutr
	Ave		+1.1								
67	18.652M	32.0	+0.2	+0.1	+0.3	+5.7	+0.0	39.0	50.0	-11.0	Neutr
	Ave		+0.7								
^	18.652M	50.5	+0.2	+0.1	+0.3	+5.7	+0.0	57.5	50.0	+7.5	Neutr
			+0.7						see average data		
69	29.979M	31.3	+0.2	+0.2	+0.5	+5.7	+0.0	39.0	50.0	-11.0	Neutr
	Ave		+1.1								
^	29.984M	42.4	+0.2	+0.2	+0.5	+5.7	+0.0	50.1	50.0	+0.1	Neutr
			+1.1						see average data		
^	29.979M	42.0	+0.2	+0.2	+0.5	+5.7	+0.0	49.7	50.0	-0.3	Neutr
			+1.1						see average data		
72	233.629k	35.3	+0.1	+0.0	+0.0	+5.7	+0.0	41.2	52.3	-11.1	Neutr
	Ave		+0.1								
^	233.628k	50.9	+0.1	+0.0	+0.0	+5.7	+0.0	56.8	52.3	+4.5	Neutr
			+0.1						see average data		
74	25.563M	30.7	+0.2	+0.2	+0.4	+5.7	+0.0	38.1	50.0	-11.9	Neutr
	Ave		+0.9								
^	25.563M	45.5	+0.2	+0.2	+0.4	+5.7	+0.0	52.9	50.0	+2.9	Neutr
			+0.9						see average data		
76	19.058M	30.7	+0.2	+0.1	+0.3	+5.7	+0.0	37.7	50.0	-12.3	Neutr
	Ave		+0.7								
^	19.058M	46.5	+0.2	+0.1	+0.3	+5.7	+0.0	53.5	50.0	+3.5	Neutr
			+0.7						see average data		
78	28.477M	29.8	+0.2	+0.2	+0.5	+5.7	+0.0	37.4	50.0	-12.6	Neutr
	Ave		+1.0								
^	28.477M	41.9	+0.2	+0.2	+0.5	+5.7	+0.0	49.5	50.0	-0.5	Neutr
			+1.0						see average data		
80	29.713M	29.6	+0.2	+0.2	+0.5	+5.7	+0.0	37.3	50.0	-12.7	Neutr
	Ave		+1.1								
^	29.713M	42.0	+0.2	+0.2	+0.5	+5.7	+0.0	49.7	50.0	-0.3	Neutr
			+1.1						see average data		
82	193.632k	34.9	+0.1	+0.0	+0.0	+5.7	+0.0	40.8	53.9	-13.1	Neutr
	Ave		+0.1								
^	193.632k	55.2	+0.1	+0.0	+0.0	+5.7	+0.0	61.1	53.9	+7.2	Neutr
			+0.1						see average data		
84	255.445k	32.5	+0.1	+0.0	+0.0	+5.7	+0.0	38.4	51.6	-13.2	Neutr
	Ave		+0.1								
^	255.444k	48.5	+0.1	+0.0	+0.0	+5.7	+0.0	54.4	51.6	+2.8	Neutr
			+0.1						see average data		
86	26.158M	29.3	+0.2	+0.2	+0.4	+5.7	+0.0	36.7	50.0	-13.3	Neutr
	Ave		+0.9								
^	26.158M	42.1	+0.2	+0.2	+0.4	+5.7	+0.0	49.5	50.0	-0.5	Neutr
			+0.9						see average data		
88	16.125M	28.8	+0.2	+0.1	+0.3	+5.7	+0.0	35.7	50.0	-14.3	Neutr
	Ave		+0.6								
^	16.125M	43.4	+0.2	+0.1	+0.3	+5.7	+0.0	50.3	50.0	+0.3	Neutr
			+0.6						see average data		
90	28.062M	28.0	+0.2	+0.2	+0.5	+5.7	+0.0	35.6	50.0	-14.4	Neutr
	Ave		+1.0								
^	28.062M	44.3	+0.2	+0.2	+0.5	+5.7	+0.0	51.9	50.0	+1.9	Neutr
			+1.0						see average data		

92	29.962M	27.3	+0.2	+0.2	+0.5	+5.7	+0.0	35.0	50.0	-15.0	Neutr
	Ave		+1.1								
^	29.962M	43.5	+0.2	+0.2	+0.5	+5.7	+0.0	51.2	50.0	+1.2	Neutr
			+1.1						see average data		
94	26.014M	27.5	+0.2	+0.2	+0.4	+5.7	+0.0	34.9	50.0	-15.1	Neutr
	Ave		+0.9								
^	26.014M	44.5	+0.2	+0.2	+0.4	+5.7	+0.0	51.9	50.0	+1.9	Neutr
			+0.9						see average data		
96	16.008M	27.9	+0.2	+0.1	+0.3	+5.7	+0.0	34.8	50.0	-15.2	Neutr
	Ave		+0.6								
^	16.008M	43.7	+0.2	+0.1	+0.3	+5.7	+0.0	50.6	50.0	+0.6	Neutr
			+0.6						see average data		
98	26.185M	27.3	+0.2	+0.2	+0.4	+5.7	+0.0	34.7	50.0	-15.3	Neutr
	Ave		+0.9								
^	26.185M	42.5	+0.2	+0.2	+0.4	+5.7	+0.0	49.9	50.0	-0.1	Neutr
			+0.9						see average data		
100	26.113M	27.1	+0.2	+0.2	+0.4	+5.7	+0.0	34.5	50.0	-15.5	Neutr
	Ave		+0.9								
^	26.113M	44.6	+0.2	+0.2	+0.4	+5.7	+0.0	52.0	50.0	+2.0	Neutr
			+0.9						see average data		
102	566.325k	22.2	+0.2	+0.0	+0.0	+5.7	+0.0	28.2	46.0	-17.8	Neutr
	Ave		+0.1								
^	566.325k	41.3	+0.2	+0.0	+0.0	+5.7	+0.0	47.3	46.0	+1.3	Neutr
			+0.1						see average data		
104	655.408k	21.8	+0.2	+0.0	+0.0	+5.7	+0.0	27.8	46.0	-18.2	Neutr
	Ave		+0.1								
^	655.408k	39.5	+0.2	+0.0	+0.0	+5.7	+0.0	45.5	46.0	-0.5	Neutr
			+0.1						see average data		
106	200.904k	28.9	+0.1	+0.0	+0.0	+5.7	+0.0	34.8	53.6	-18.8	Neutr
	Ave		+0.1								
^	200.904k	50.7	+0.1	+0.0	+0.0	+5.7	+0.0	56.6	53.6	+3.0	Neutr
			+0.1						see average data		
108	268.171k	25.4	+0.1	+0.0	+0.0	+5.7	+0.0	31.3	51.2	-19.9	Neutr
	Ave		+0.1								
^	268.170k	48.9	+0.1	+0.0	+0.0	+5.7	+0.0	54.8	51.2	+3.6	Neutr
			+0.1						see average data		

Test Setup Photo(s)



15.109 Radiated Emissions

Test Notes: Radiated disturbances emanating from enclosure.

Test Setup / Conditions / Data

Test Location:	CKC Laboratories, Inc. • 110 N. Olinda Place • Brea, CA • 714-993-6112		
Customer:	**Multisensory Fitness, Inc.**		
Specification:	**15.109 Radiated Emissions Class B**		
Work Order #:	**97844**	Date:	7/8/2016
Test Type:	**Maximized Emissions**	Time:	14:31:26
Tested by:	S. Yamamoto	Sequence#:	2
Software:	EMITest 5.03.02		

Equipment Tested:

Device	Manufacturer	Model #	S/N
Configuration 1			

Support Equipment:

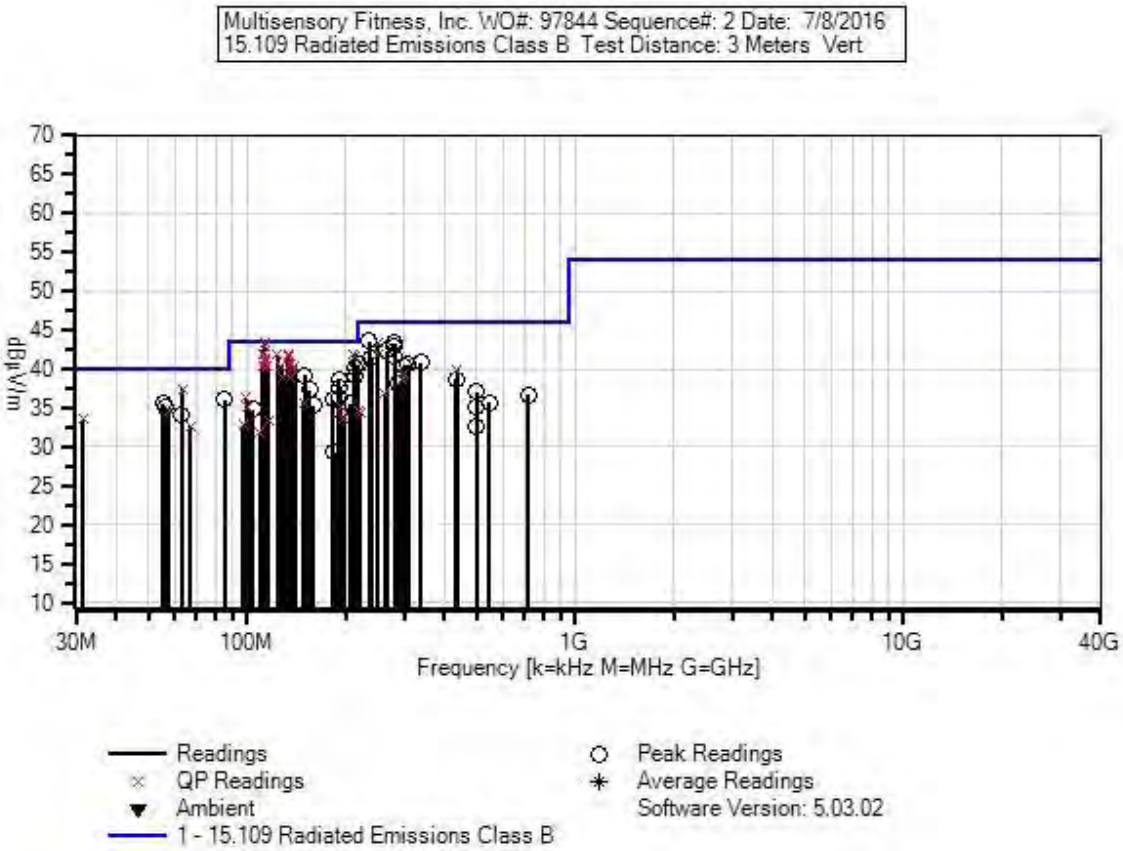
Device	Manufacturer	Model #	S/N
Configuration 1			

Test Conditions / Notes:

The equipment under test (EUT) is placed on the open area test site. The EUT is in its normal mode of operation set to game 600. Frequency range of data sheet is 30MHz to 1000MHz. RBW=VBW=120kHz.

Temperature: 26°
Relative Humidity: 52%
Atmospheric Pressure: 100kPa

Site D.
The manufacturer declares that the highest frequency generated or used in the device is less than 108MHz.
Voltage to the EUT is 120Vac 60Hz.
Test Method: ANSI C63.4 (2014)
Modification 1 was in place during testing.


Multisensory Fitness, Inc. WO#: 97844 Sequence#: 2 Date: 7/8/2016
15.109 Radiated Emissions Class B Test Distance: 3 Meters Vert
dBµV/m
70
65
60
55
50
45
40
35
30
25
20
15
10
30M
100M
1G
10G
40G
Frequency [k=kHz M=MHz G=GHz]
Readings
QP Readings
Ambient
1 - 15.109 Radiated Emissions Class B
Peak Readings
Average Readings
Software Version: 5.03.02

Test Equipment:

ID	Asset	Description	Model	Calibration Date	Cal Due Date
T1	AN02467	Spectrum Analyzer	E7405A	5/10/2016	5/10/2017
T2	ANP04382	Cable	LDF-50	6/6/2016	6/6/2018
T3	ANP05555	Cable	RG223/U	4/5/2016	4/5/2018
T4	ANP05569	Cable	RG-214/U	4/4/2016	4/4/2018
T5	AN00010	Preamp	8447D	3/14/2016	3/14/2018
T6	AN01992	Biconilog Antenna	CBL6111C	12/4/2014	12/4/2016
T7	ANP05283	Attenuator	ATT-0218-06-NNN-02	5/5/2016	5/5/2018

Measurement Data: Reading listed by margin. Test Distance: 3 Meters

#	Freq	Rdng	T1 T5	T2 T6	T3 T7	T4	Dist	Corr	Spec	Margin	Polar
	MHz	dBµV	dB	dB	dB	dB	Table	dBµV/m	dBµV/m	dB	Ant
1	113.235M	51.1	+0.0	+1.1	+0.1	+1.1	+0.0	43.3	43.5	-0.2	Vert
	QP		-27.0	+11.1	+5.8						
^	113.235M	53.3	+0.0	+1.1	+0.1	+1.1	+0.0	45.5	43.5	+2.0	Vert
			-27.0	+11.1	+5.8						
3	112.970M	50.6	+0.0	+1.1	+0.1	+1.1	+0.0	42.8	43.5	-0.7	Vert
	QP		-27.0	+11.1	+5.8						
^	112.970M	51.3	+0.0	+1.1	+0.1	+1.1	+0.0	43.5	43.5	+0.0	Vert
			-27.0	+11.1	+5.8						
5	133.260M	49.3	+0.0	+1.2	+0.1	+1.2	+0.0	42.0	43.5	-1.5	Horiz
	QP		-27.0	+11.4	+5.8						
^	133.260M	53.0	+0.0	+1.2	+0.1	+1.2	+0.0	45.7	43.5	+2.2	Horiz
			-27.0	+11.4	+5.8						
7	112.170M	49.9	+0.0	+1.1	+0.1	+1.1	+0.0	42.0	43.5	-1.5	Vert
	QP		-27.0	+11.0	+5.8						
^	112.170M	51.7	+0.0	+1.1	+0.1	+1.1	+0.0	43.8	43.5	+0.3	Vert
			-27.0	+11.0	+5.8						
9	211.350M	49.7	+0.0	+1.5	+0.2	+1.5	+0.0	42.0	43.5	-1.5	Horiz
	QP		-26.7	+10.0	+5.8						
^	211.350M	51.4	+0.0	+1.5	+0.2	+1.5	+0.0	43.7	43.5	+0.2	Horiz
			-26.7	+10.0	+5.8						
11	123.600M	49.2	+0.0	+1.1	+0.1	+1.1	+0.0	41.8	43.5	-1.7	Vert
	QP		-27.0	+11.5	+5.8						
^	123.600M	55.1	+0.0	+1.1	+0.1	+1.1	+0.0	47.7	43.5	+4.2	Vert
			-27.0	+11.5	+5.8						
13	211.552M	49.3	+0.0	+1.5	+0.2	+1.5	+0.0	41.6	43.5	-1.9	Horiz
	QP		-26.7	+10.0	+5.8						
^	211.552M	53.8	+0.0	+1.5	+0.2	+1.5	+0.0	46.1	43.5	+2.6	Horiz
			-26.7	+10.0	+5.8						
15	133.677M	48.9	+0.0	+1.2	+0.1	+1.2	+0.0	41.6	43.5	-1.9	Vert
	QP		-27.0	+11.4	+5.8						
^	133.677M	54.5	+0.0	+1.2	+0.1	+1.2	+0.0	47.2	43.5	+3.7	Vert
			-27.0	+11.4	+5.8						

17	113.540M QP	49.3	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	41.5	43.5	-2.0	Vert
18	113.450M QP	48.9	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	41.1	43.5	-2.4	Vert
^	113.540M	53.2	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	45.4	43.5	+1.9	Vert
20	235.939M	49.3	+0.0 -26.7	+1.6 +11.8	+0.2 +5.8	+1.6	+0.0	43.6	46.0	-2.4	Horiz
21	250.686M QP	48.0	+0.0 -26.6	+1.6 +12.7	+0.2 +5.8	+1.7	+0.0	43.4	46.0	-2.6	Vert
^	250.686M	49.8	+0.0 -26.6	+1.6 +12.7	+0.2 +5.8	+1.7	+0.0	45.2	46.0	-0.8	Vert
^	250.672M	47.4	+0.0 -26.6	+1.6 +12.7	+0.2 +5.8	+1.7	+0.0	42.8	46.0	-3.2	Vert
24	63.550M QP	51.0	+0.0 -27.2	+0.8 +6.2	+0.1 +5.8	+0.7	+0.0	37.4	40.0	-2.6	Vert
^	63.550M	54.4	+0.0 -27.2	+0.8 +6.2	+0.1 +5.8	+0.7	+0.0	40.8	40.0	+0.8	Vert
26	280.190M	47.3	+0.0 -26.6	+1.7 +13.1	+0.2 +5.8	+1.8	+0.0	43.3	46.0	-2.7	Horiz
27	113.405M QP	48.6	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	40.8	43.5	-2.7	Vert
^	113.405M	54.9	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	47.1	43.5	+3.6	Vert
^	113.450M	54.1	+0.0 -27.0	+1.1 +11.1	+0.1 +5.8	+1.1	+0.0	46.3	43.5	+2.8	Vert
30	135.360M QP	47.8	+0.0 -27.0	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.5	43.5	-3.0	Horiz
31	139.560M QP	47.7	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.5	43.5	-3.0	Vert
^	139.560M	49.2	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	42.0	43.5	-1.5	Vert
33	127.040M QP	47.8	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	40.4	43.5	-3.1	Horiz
^	127.040M	52.7	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	45.3	43.5	+1.8	Horiz
35	280.169M	46.9	+0.0 -26.6	+1.7 +13.1	+0.2 +5.8	+1.8	+0.0	42.9	46.0	-3.1	Horiz
36	115.020M QP	48.0	+0.0 -27.0	+1.1 +11.2	+0.1 +5.8	+1.1	+0.0	40.3	43.5	-3.2	Vert
^	115.020M	51.8	+0.0 -27.0	+1.1 +11.2	+0.1 +5.8	+1.1	+0.0	44.1	43.5	+0.6	Vert
38	111.645M QP	48.3	+0.0 -27.1	+1.1 +11.0	+0.1 +5.8	+1.1	+0.0	40.3	43.5	-3.2	Vert
^	111.645M	53.1	+0.0 -27.1	+1.1 +11.0	+0.1 +5.8	+1.1	+0.0	45.1	43.5	+1.6	Vert
40	138.085M QP	47.4	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	40.2	43.5	-3.3	Vert
^	138.085M	52.4	+0.0 -26.9	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	45.2	43.5	+1.7	Vert

42	213.008M	47.7	+0.0 -26.7	+1.5 +10.1	+0.2 +5.8	+1.5	+0.0	40.1	43.5	-3.4	Horiz
43	265.445M	46.7	+0.0 -26.6	+1.7 +12.9	+0.2 +5.8	+1.7	+0.0	42.4	46.0	-3.6	Horiz
44	85.050M	47.0	+0.0 -27.1	+1.0 +8.4	+0.1 +5.8	+0.9	+0.0	36.1	40.0	-3.9	Vert
45	250.702M QP	46.4	+0.0 -26.6	+1.6 +12.7	+0.2 +5.8	+1.7	+0.0	41.8	46.0	-4.2	Horiz
^	250.702M	49.0	+0.0 -26.6	+1.6 +12.7	+0.2 +5.8	+1.7	+0.0	44.4	46.0	-1.6	Horiz
47	149.675M	46.3	+0.0 -26.9	+1.3 +11.2	+0.2 +5.8	+1.3	+0.0	39.2	43.5	-4.3	Vert
48	212.837M	46.8	+0.0 -26.7	+1.5 +10.1	+0.2 +5.8	+1.5	+0.0	39.2	43.5	-4.3	Horiz
49	280.199M	45.7	+0.0 -26.6	+1.7 +13.1	+0.2 +5.8	+1.8	+0.0	41.7	46.0	-4.3	Horiz
50	55.300M	48.1	+0.0 -27.2	+0.8 +7.3	+0.1 +5.8	+0.7	+0.0	35.6	40.0	-4.4	Vert
51	238.100M	47.0	+0.0 -26.6	+1.6 +11.9	+0.2 +5.8	+1.6	+0.0	41.5	46.0	-4.5	Horiz
52	128.300M QP	46.3	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	38.9	43.5	-4.6	Horiz
^	128.300M	51.8	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	44.4	43.5	+0.9	Horiz
^	128.300M	51.5	+0.0 -27.0	+1.1 +11.5	+0.1 +5.8	+1.1	+0.0	44.1	43.5	+0.6	Horiz
55	135.400M QP	46.1	+0.0 -27.0	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	38.8	43.5	-4.7	Horiz
^	135.360M	54.5	+0.0 -27.0	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	47.2	43.5	+3.7	Horiz
^	135.400M	50.8	+0.0 -27.0	+1.2 +11.4	+0.1 +5.8	+1.2	+0.0	43.5	43.5	+0.0	Horiz
58	190.075M	47.5	+0.0 -26.8	+1.5 +9.1	+0.2 +5.8	+1.4	+0.0	38.7	43.5	-4.8	Vert
59	56.287M	47.8	+0.0 -27.2	+0.8 +7.1	+0.1 +5.8	+0.7	+0.0	35.1	40.0	-4.9	Vert
60	339.157M	42.9	+0.0 -26.8	+2.0 +14.6	+0.3 +5.8	+2.0	+0.0	40.8	46.0	-5.2	Horiz
61	57.613M QP	47.7	+0.0 -27.2	+0.8 +6.8	+0.1 +5.8	+0.7	+0.0	34.7	40.0	-5.3	Horiz
^	57.613M	48.8	+0.0 -27.2	+0.8 +6.8	+0.1 +5.8	+0.7	+0.0	35.8	40.0	-4.2	Horiz
63	309.669M	44.1	+0.0 -26.7	+1.8 +13.6	+0.2 +5.8	+1.8	+0.0	40.6	46.0	-5.4	Horiz
64	217.475M	47.8	+0.0 -26.7	+1.5 +10.5	+0.2 +5.8	+1.5	+0.0	40.6	46.0	-5.4	Vert
65	300.710M	44.2	+0.0 -26.6	+1.8 +13.3	+0.2 +5.8	+1.8	+0.0	40.5	46.0	-5.5	Horiz
66	192.075M	46.4	+0.0 -26.8	+1.5 +9.1	+0.2 +5.8	+1.4	+0.0	37.6	43.5	-5.9	Vert
67	63.088M	47.6	+0.0 -27.2	+0.8 +6.2	+0.1 +5.8	+0.7	+0.0	34.0	40.0	-6.0	Horiz

68	436.000M	40.0	+0.0	+2.2	+0.3	+2.3	+0.0	40.0	46.0	-6.0	Horiz
	QP		-27.6	+17.0	+5.8						
^	436.000M	44.6	+0.0	+2.2	+0.3	+2.3	+0.0	44.6	46.0	-1.4	Horiz
			-27.6	+17.0	+5.8						
70	154.485M	44.6	+0.0	+1.3	+0.2	+1.3	+0.0	37.3	43.5	-6.2	Horiz
			-26.9	+11.0	+5.8						
71	31.650M	36.4	+0.0	+0.5	+0.0	+0.5	+0.0	33.5	40.0	-6.5	Vert
	QP		-27.2	+17.5	+5.8						
^	31.650M	40.3	+0.0	+0.5	+0.0	+0.5	+0.0	37.4	40.0	-2.6	Vert
			-27.2	+17.5	+5.8						
73	301.360M	43.1	+0.0	+1.8	+0.2	+1.8	+0.0	39.4	46.0	-6.6	Horiz
	QP		-26.6	+13.3	+5.8						
^	301.360M	48.9	+0.0	+1.8	+0.2	+1.8	+0.0	45.2	46.0	-0.8	Horiz
			-26.6	+13.3	+5.8						
75	99.278M	45.4	+0.0	+1.0	+0.1	+1.0	+0.0	36.3	43.5	-7.2	Vert
	QP		-27.1	+10.1	+5.8						
^	99.278M	48.3	+0.0	+1.0	+0.1	+1.0	+0.0	39.2	43.5	-4.3	Vert
			-27.1	+10.1	+5.8						
77	191.675M	45.1	+0.0	+1.5	+0.2	+1.4	+0.0	36.3	43.5	-7.2	Vert
			-26.8	+9.1	+5.8						
78	67.250M	46.3	+0.0	+0.8	+0.1	+0.8	+0.0	32.7	40.0	-7.3	Vert
	QP		-27.2	+6.1	+5.8						
^	67.250M	50.7	+0.0	+0.8	+0.1	+0.8	+0.0	37.1	40.0	-2.9	Vert
			-27.2	+6.1	+5.8						
80	434.000M	38.8	+0.0	+2.2	+0.3	+2.3	+0.0	38.7	46.0	-7.3	Vert
			-27.6	+16.9	+5.8						
81	294.919M	42.4	+0.0	+1.8	+0.2	+1.8	+0.0	38.6	46.0	-7.4	Horiz
	QP		-26.6	+13.2	+5.8						
^	294.928M	47.2	+0.0	+1.8	+0.2	+1.8	+0.0	43.4	46.0	-2.6	Horiz
			-26.6	+13.2	+5.8						
^	294.919M	46.0	+0.0	+1.8	+0.2	+1.8	+0.0	42.2	46.0	-3.8	Horiz
			-26.6	+13.2	+5.8						
84	182.350M	44.9	+0.0	+1.4	+0.2	+1.4	+0.0	36.0	43.5	-7.5	Vert
			-26.8	+9.1	+5.8						
85	148.350M	43.1	+0.0	+1.2	+0.1	+1.2	+0.0	35.7	43.5	-7.8	Vert
	QP		-26.9	+11.2	+5.8						
^	148.350M	48.6	+0.0	+1.2	+0.1	+1.2	+0.0	41.2	43.5	-2.3	Vert
			-26.9	+11.2	+5.8						
87	288.000M	41.9	+0.0	+1.8	+0.2	+1.8	+0.0	38.1	46.0	-7.9	Vert
			-26.6	+13.2	+5.8						
88	206.453M	43.7	+0.0	+1.5	+0.2	+1.4	+0.0	35.5	43.5	-8.0	Horiz
			-26.7	+9.6	+5.8						
89	159.610M	42.8	+0.0	+1.3	+0.2	+1.3	+0.0	35.3	43.5	-8.2	Horiz
			-26.9	+10.8	+5.8						
90	100.570M	44.0	+0.0	+1.0	+0.1	+1.0	+0.0	35.0	43.5	-8.5	Vert
	QP		-27.1	+10.2	+5.8						
^	100.570M	49.5	+0.0	+1.0	+0.1	+1.0	+0.0	40.5	43.5	-3.0	Vert
			-27.1	+10.2	+5.8						
92	192.370M	43.8	+0.0	+1.5	+0.2	+1.4	+0.0	35.0	43.5	-8.5	Vert
	QP		-26.8	+9.1	+5.8						
^	192.370M	49.8	+0.0	+1.5	+0.2	+1.4	+0.0	41.0	43.5	-2.5	Vert
			-26.8	+9.1	+5.8						

94	104.009M	43.5	+0.0 -27.1	+1.0 +10.5	+0.1 +5.8	+1.0	+0.0	34.8	43.5	-8.7	Vert
95	191.704M	43.4	+0.0 -26.8	+1.5 +9.1	+0.2 +5.8	+1.4	+0.0	34.6	43.5	-8.9	Horiz
96	501.379M	35.9	+0.0 -28.0	+2.4 +18.0	+0.4 +5.8	+2.5	+0.0	37.0	46.0	-9.0	Horiz
97	262.177M QP	41.1	+0.0 -26.6	+1.7 +12.9	+0.2 +5.8	+1.7	+0.0	36.8	46.0	-9.2	Horiz
^	262.177M	47.7	+0.0 -26.6	+1.7 +12.9	+0.2 +5.8	+1.7	+0.0	43.4	46.0	-2.6	Horiz
99	720.101M	31.8	+0.0 -28.0	+2.9 +20.7	+0.4 +5.8	+3.1	+0.0	36.7	46.0	-9.3	Horiz
100	197.320M QP	42.4	+0.0 -26.7	+1.5 +9.1	+0.2 +5.8	+1.4	+0.0	33.7	43.5	-9.8	Vert
^	197.320M	47.6	+0.0 -26.7	+1.5 +9.1	+0.2 +5.8	+1.4	+0.0	38.9	43.5	-4.6	Vert
102	116.338M QP	41.0	+0.0 -27.0	+1.1 +11.3	+0.1 +5.8	+1.1	+0.0	33.4	43.5	-10.1	Horiz
^	116.338M	47.3	+0.0 -27.0	+1.1 +11.3	+0.1 +5.8	+1.1	+0.0	39.7	43.5	-3.8	Horiz
104	545.592M	33.6	+0.0 -28.1	+2.5 +18.9	+0.3 +5.8	+2.6	+0.0	35.6	46.0	-10.4	Horiz
105	97.296M QP	42.1	+0.0 -27.1	+1.0 +9.9	+0.1 +5.8	+1.0	+0.0	32.8	43.5	-10.7	Vert
^	97.296M	47.1	+0.0 -27.1	+1.0 +9.9	+0.1 +5.8	+1.0	+0.0	37.8	43.5	-5.7	Vert
107	501.372M	34.1	+0.0 -28.0	+2.4 +18.0	+0.4 +5.8	+2.5	+0.0	35.2	46.0	-10.8	Vert
108	220.050M QP	41.7	+0.0 -26.7	+1.5 +10.6	+0.2 +5.8	+1.5	+0.0	34.6	46.0	-11.4	Horiz
^	220.050M	46.2	+0.0 -26.7	+1.5 +10.6	+0.2 +5.8	+1.5	+0.0	39.1	46.0	-6.9	Horiz
110	109.000M QP	40.3	+0.0 -27.1	+1.0 +10.8	+0.1 +5.8	+1.0	+0.0	31.9	43.5	-11.6	Vert
^	109.000M	47.2	+0.0 -27.1	+1.0 +10.8	+0.1 +5.8	+1.0	+0.0	38.8	43.5	-4.7	Vert
112	221.203M QP	41.4	+0.0 -26.7	+1.5 +10.7	+0.2 +5.8	+1.5	+0.0	34.4	46.0	-11.6	Horiz
^	221.203M	43.5	+0.0 -26.7	+1.5 +10.7	+0.2 +5.8	+1.5	+0.0	36.5	46.0	-9.5	Horiz
114	501.375M	31.4	+0.0 -28.0	+2.4 +18.0	+0.4 +5.8	+2.5	+0.0	32.5	46.0	-13.5	Horiz
115	183.360M	38.1	+0.0 -26.8	+1.4 +9.1	+0.2 +5.8	+1.4	+0.0	29.2	43.5	-14.3	Horiz

Test Setup Photo(s)





SUPPLEMENTAL INFORMATION

Measurement Uncertainty

Uncertainty Value	Parameter
4.73 dB	Radiated Emissions
3.34 dB	Mains Conducted Emissions
3.30 dB	Disturbance Power

Reported uncertainties represent expanded uncertainties expressed at approximately the 95% confidence level using a coverage factor of k=2.

Emissions Test Details

TESTING PARAMETERS

Unless otherwise indicated, the following configuration parameters are used for equipment setup: The cables were routed consistent with the typical application by varying the configuration of the test sample. Interface cables were connected to the available ports of the test unit. The effect of varying the position of the cables was investigated to find the configuration that produced maximum emissions. Cables were of the type and length specified in the individual requirements. The length of cable that produced maximum emissions was selected.

The equipment under test (EUT) was set up in a manner that represented its normal use, as shown in the setup photographs. Any special conditions required for the EUT to operate normally are identified in the comments that accompany the emissions tables.

The emissions data was taken with a spectrum analyzer or receiver. Incorporating the applicable correction factors for distance, antenna, cable loss and amplifier gain, the data was reduced as shown in the table below. The corrected data was then compared to the applicable emission limits. Preliminary and final measurements were taken in order to ensure that all emissions from the EUT were found and maximized.

CORRECTION FACTORS

The basic spectrum analyzer reading was converted using correction factors as shown in the highest emissions readings in the tables. For radiated emissions in dBµV/m, the spectrum analyzer reading in dBµV was corrected by using the following formula. This reading was then compared to the applicable specification limit. Individual measurements were compared with the displayed limit value in the margin column. The margin was calculated based on subtracting the limit value from the corrected measurement value; a positive margin represents a measurement exceeding the limit, while a negative margin represents a measurement less than the limit.

SAMPLE CALCULATIONS		
	Meter reading	(dBµV)
+	Antenna Factor	(dB/m)
+	Cable Loss	(dB)
-	Distance Correction	(dB)
-	Preamplifier Gain	(dB)
=	Corrected Reading	(dBµV/m)

TEST INSTRUMENTATION AND ANALYZER SETTINGS

The test instrumentation and equipment listed were used to collect the emissions data. A spectrum analyzer or receiver was used for all measurements. Unless otherwise specified, the following table shows the measuring equipment bandwidth settings that were used in designated frequency bands. For testing emissions, an appropriate reference level and a vertical scale size of 10 dB per division were used.

MEASURING EQUIPMENT BANDWIDTH SETTINGS PER FREQUENCY RANGE			
TEST	BEGINNING FREQUENCY	ENDING FREQUENCY	BANDWIDTH SETTING
CONDUCTED EMISSIONS	150 kHz	30 MHz	9 kHz
RADIATED EMISSIONS	9 kHz	150 kHz	200 Hz
RADIATED EMISSIONS	150 kHz	30 MHz	9 kHz
RADIATED EMISSIONS	30 MHz	1000 MHz	120 kHz
RADIATED EMISSIONS	1000 MHz	>1 GHz	1 MHz

SPECTRUM ANALYZER/RECEIVER DETECTOR FUNCTIONS

The notes that accompany the measurements contained in the emissions tables indicate the type of detector function used to obtain the given readings. Unless otherwise noted, all readings were made in the "positive peak" detector mode. Whenever a "quasi-peak" or "average" reading was recorded, the measurement was annotated with a "QP" or an "Ave" on the appropriate rows of the data sheets. In cases where quasi-peak or average limits were employed and data exists for multiple measurement types for the same frequency then the peak measurement was retained in the report for reference, however the numbering for the affected row was removed and an arrow or caret ("^") was placed in the far left-hand column indicating that the row above takes precedence for comparison to the limit. The following paragraphs describe in more detail the detector functions and when they were used to obtain the emissions data.

Peak

In this mode, the spectrum analyzer or receiver recorded all emissions at their peak value as the frequency band selected was scanned. By combining this function with another feature called "peak hold," the measurement device had the ability to measure intermittent or low duty cycle transient emission peak levels. In this mode the measuring device made a slow scan across the frequency band selected and measured the peak emission value found at each frequency across the band.

Quasi-Peak

Quasi-peak measurements were taken using the quasi-peak detector when the true peak values exceeded or were within 2 dB of a quasi-peak specification limit. Additional QP measurements may have been taken at the discretion of the operator.

Average

Average measurements were taken using the average detector when the true peak values exceeded or were within 2 dB of an average specification limit. Additional average measurements may have been taken at the discretion of the operator. If the specification or test procedure requires trace averaging, then the averaging was performed using 100 samples or as required by the specification. All other average measurements are performed using video bandwidth averaging. To make these measurements, the test engineer reduces the video bandwidth on the measuring device until the modulation of the signal is filtered out. At this point the measuring device is set into the linear mode and the scan time is reduced.